

SOUMIK SARKER ANIK

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SUMMARY

Master's graduate in Software Engineering with almost **three years** of research and industry experience in the field of machine learning and data science. Highly proficient in **deep learning models** and fuzzy logic systems. Co-author of **three journal articles** and two conference papers with a focus on predictive modeling and the analysis of complex geospatial, environmental, healthcare, and time series datasets. Fast learner, self-motivated, and tech enthusiast with excellent analytical and coding skills proven through work experience.

EDUCATION

Memorial University of Newfoundland
Masters of Applied Science, Software Engineering

2023- 2025
St John's, NL

North South University
Bachelor of Science, Computer Science - GPA 3.77/4.00 (Magna Cum Laude)

2017-2021
Dhaka, Bangladesh

WORK EXPERIENCE

Online Data Analyst (Freelancer)

- TELUS Digital** 05/2025-Present
- Evaluating AI-generated content for accuracy, factual correctness, groundedness, and adherence to composition standards using internal guidelines.
 - Categorizing and annotating** search queries based on user intent to support large-scale supervised ML model training.
 - Providing qualitative feedback to improve AI response logic and enhance model outputs for real-world user experience.

Machine Learning Engineer (Intern)

- ExxonMobil Corporation** 09/2024-12/2024
- Developed and implemented **unsupervised ML models** for anomaly detection in industrial machinery time series data.
 - Performed necessary feature engineering and preprocessing steps on the data to enhance model performance
 - Utilized Azure Databricks and Azure Data Explorer (ADX) for data extraction and contributed to designing the solution architecture.

University Research Assistant

- North South University** 09/2019-02/2022
- Published three journal articles and two book chapters showcasing expertise in advanced ML algorithms across domains, with over **65 citations**.
 - Implemented different classification, clustering, and hybrid neural network models with my research team of five people
 - Conducted research in diverse domains, including healthcare analytics (COVID-19 transmission modeling), geospatial analysis (flood risk assessment), and transportation safety (accident severity prediction).

University Teaching Assistant

- North South University** 05/2019-10/2020
- Assisted respective faculty members in preparing lesson plans, graded assignments, and quiz scripts for each section
 - Gave one-to-one consultations to over 30 students about the topics covered in classes

PROJECTS AND PUBLICATIONS

- Time Series Anomaly Detection in Gas Compression Data (Project)**
Developed unsupervised models (DBSCAN, Isolation Forest) to identify equipment anomalies using time series sensor data.
- Flood Risk Assessment in Bangladesh (Publication - Geocarto International, 2022)**
GIS-based national-scale flood risk modeling using hybrid DNN and fuzzy AHP methods.
- Hybridized SVR for Flood Susceptibility Mapping (Publication - IEA/AIE 2021)**
Applied Genetic Algorithm-optimized SVR models for flood susceptibility prediction using GIS data.
- Fuzzy SEIRD for COVID-19 Analysis (Publication - Int. J. of Modern Physics C, 2022)**
Enhanced SEIRD model using fuzzy logic to simulate SARS-CoV-2 transmission dynamics.
- Linguistic Fuzzy Rule-Based COVID-19 Modeling (Publication - Int. J. of Fuzzy System Applications, 2022)**
Developed theoretical compartmental models using fuzzy linguistic rules for pandemic prediction.
- Road Crash Severity Prediction Using Machine Learning (Publication - ICCCI 2020)**
Machine learning-based approach using AHC, Random Forest, and C5.0 models for crash severity prediction.

SKILLS

- Programming Languages: Python, R, SQL, C/C++, MATLAB, GIS Softwares (ArcGIS)
- Machine learning and NLP: scikit-learn, keras, tensorflow, pytorch, NLTK
- Data Visualizations: Matplotlib, Seaborn, PowerBI
- MLOps and DevOps: CI/CD pipelines, MLflow, version control(Git), Azure Databricks, Azure Data Explorer (ADX)
- Domain Specific: Healthcare Data Analytics and Modelling, Geospatial Data Analysis, Hybrid ML models, Time series Data Analysis, Predictive Modelling, Anomaly Detection